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import java.util.*;

public class Afine_Cypher
{
    public static int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }
}
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public static String Encryption() {
    Scanner sc = new Scanner(System.in);
    int key;
    char[] text;

    while (true) {
        System.out.print("Enter the Text in LowerCase : ");
        String str = sc.nextLine();
        text = str.toCharArray();
        boolean valid = true;

        for (int i = 0; i < text.length; i++) {
            if (text[i] == ' ') {
                continue;
            }
            else if (text[i] < 'a' || text[i] > 'z') {
                valid = false;
                break;
            }
        }

        if (!valid) {
            System.out.println(
                "Please Enter text only and also make sure it is in lower Case:"
            );
            continue;
        }
        else {
            break;
        }
    }

    while (true) {
        System.out.print("Enter the Key : ");

        try {
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        key = sc.nextInt();

        if (gcd(key, 26) == 1 && key < 26 && key >= 1) {
            break;
        }

        System.out.println("Invalid Key");
    }
    catch (InputMismatchException e) {
        System.out.println("Invalid input! Please enter an integer.");
        sc.next();
    }
}

char cypher[] = new char[text.length];

for (int i = 0; i < text.length; i++) {
    if (text[i] == ' ') {
        cypher[i] = text[i];
        continue;
    }

    char a = text[i];

    int n = (int) a - 97;
    int t = ((n * key) % 26);

    cypher[i] = (char) (t + 97);
}

int key2;

while (true) {
    System.out.print("Enter The Second Key: ");

    try {
        key2 = sc.nextInt();

        if (key2 <= 26 && key2 >= 1) {
            break;
        }

        System.out.println("Invalid Key");
    }
    catch (InputMismatchException e) {
        System.out.println("Invalid input! Please enter an integer.");
        sc.next();
    }
}

char cypher2[] = new char[cypher.length];

for (int i = 0; i < cypher.length; i++) {

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        if (cypher[i] == ' ') {
            cypher[i] = text[i];
            continue;
        }

        int a = cypher[i] - 97;
        int x = (a + key2) % 26;

        char y = (char) (x + 97);
        cypher2[i] = y;
    }

    String cyphertext = new String(cypher2);
    cyphertext = cyphertext.toUpperCase();

    return cyphertext;
}

public static String Decryption() {
    Scanner sc = new Scanner(System.in);
    int key2;

    char[] text;

    System.out.print("Enter The Encrypted text In UpperCase: ");
    String str = sc.nextLine();

    str = str.toLowerCase();
    text = str.toCharArray();

    char text2[] = new char[text.length];

    while (true) {
        System.out.print("Enter The Second Key: ");

        try {
            key2 = sc.nextInt();

            if (key2 <= 26 && key2 >= 1) {
                break;
            }

            System.out.println("Invalid Key");
        }
        catch (InputMismatchException e) {
            System.out.println("Invalid input! Please enter an integer.");
            sc.next();
        }
    }

    for (int i = 0; i < text.length; i++) {
        if (text[i] == ' ') {
            text2[i] = text[i];

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        continue;
    }

    int a = text[i] - 97;
    int x = (a - key2 + 26) % 26;

    char y = (char) (x + 97);
    text2[i] = y;
}

int key;

while (true) {
    System.out.print("Enter First Key Used In Encryption: ");

    try {
        key = sc.nextInt();

        if (gcd(key, 26) == 1 && key < 26 && key >= 1) {
            break;
        }

        System.out.println("Invalid Key");
    }
    catch (InputMismatchException e) {
        System.out.println("Invalid input! Please enter an integer.");
        sc.next();
    }
}

char cypher[] = new char[text.length];

for (int i = 0; i < text2.length; i++) {
    if (text2[i] == ' ') {
        cypher[i] = ' ';
        continue;
    }

    char a = text2[i];

    int n = (int) a - 97;
    int na = 0;

    for (int j = 1; j <= 26; j = j + 2) {
        if (j == 13) {
            continue;
        }

        if ((key * j) % 26 == 1) {
            na = j;
            break;
        }
    }
}

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        int t = ((n * na) % 26);

        cypher[i] = (char) (t + 97);
    }

    String cyphertext = new String(cypher);
    cyphertext = cyphertext.toLowerCase();

    return cyphertext;
}

public static void Brute_force(String x, String y) {

    for (int m = 1; m <= 26; m++) {

        char e[] = x.toCharArray();
        char c[] = new char[e.length];
        char cy[] = new char[e.length];

        for (int i = 0; i < e.length; i++) {

            if (e[i] == ' ') {
                c[i] = e[i];
                continue;
            }

            int a = e[i] - 97;
            int v = (a - m + 26) % 26;

            char z = (char) (v + 97);
            c[i] = z;
        }

        int key = -1;

        for (int k = 1; k < 26; k = k + 2) {

            if (k == 13) {
                continue;
            }

            for (int i = 0; i < c.length; i++) {

                if (c[i] == ' ') {
                    cy[i] = ' ';
                    continue;
                }

                char a = c[i];

                int n = (int) a - 97;
                int na = 0;
            }
        }
    }
}

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        for (int j = 1; j <= 26; j = j + 2) {

            if (j == 13) {
                continue;
            }

            if ((k * j) % 26 == 1) {
                na = j;
                break;
            }
        }

        int t = ((n * na) % 26);

        cy[i] = (char) (t + 97);
    }

    String s = new String(cy);

    if (y.equals(s)) {
        System.out.println(
            "Second Key or Multiplicative is : " + m
        );

        System.out.println(
            "First Key or Additive is : " + k
        );

        return;
    }
}

return;
}

public static void main(String s[]) {

    while (true) {

        Scanner sc = new Scanner(System.in);

        System.out.print(
            "For Encryption : 1 " +
            "\t For Decryption : 2" +
            "\nFor Bruteforce : 3" +
            "\t For Exit : 99" +
            "\nEnter Your Choice : "
        );

        int a = sc.nextInt();
    }
}

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    if (a == 1) {
        System.out.println(
            "This is The Encrypted Text : " + Encryption()
        );
    }
    else if (a == 2) {
        System.out.println(
            "This is Decrypted Text : " + Decryption()
        );
    }
    else if (a == 3) {

        System.out.print("So First Give the Encrypted Key: ");
        sc.nextLine();

        String x = sc.nextLine();

        System.out.print("Enter The Decrypted Code :");
        String y = sc.nextLine();

        x = x.toLowerCase();

        Brute_force(x, y);
    }
    else if (a == 99) {
        break;
    }
}
}

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}

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